



3.3 Differentiating Inverse Functions

Definitions of Inverse Functions

Two functions are inverse functions when

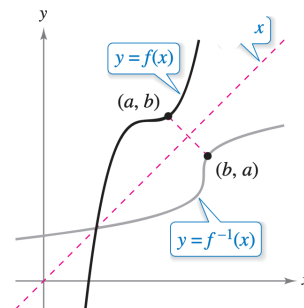
$$(f \circ f^{-1})(x) = x$$

AND

$$(f^{-1} \circ f)(x) = x$$

Inverse Properties

- The graphs of f and f^{-1} are reflections about the line $y = x$.
- The domain of f is the range of f^{-1} .
- The range of f is the domain of f^{-1} .
- A function has an inverse function if and only if it is one-to-one, that is its monotonically increasing or decreasing.



Ex 1: Find the indicated value.

a) $f(x) = x^{5/3}$, $f^{-1}(243) =$

$$\begin{aligned} x^{5/3} &= 243 \\ x &= 243^{3/5} \\ x &= 27 \\ f(27) &= 243 \Rightarrow f^{-1}(243) = 27 \end{aligned}$$

b) $f(x) = 2x + 7$, $f^{-1}(14) =$

$$\begin{aligned} 2x + 7 &= 14 \\ x &= \frac{7}{2} \\ f\left(\frac{7}{2}\right) &= 14 \Rightarrow f^{-1}(14) = \frac{7}{2} \end{aligned}$$

Discovering Relationships

Ex 2: Open the [GeoGebra applet](#). Move slider a and observe the values of f' and g' . What do you notice about the slopes $f'(a)$ and $g'(b)$?



$f'(a)$ and $g'(b)$ have reciprocal values.
(discover why in the proof below!)

The Derivative of an Inverse Function

Let f be a function that is differentiable on an interval I and has an inverse function f^{-1} . If f is differentiable at on an interval containing $x = c$ and $f'(c) \neq 0$, then f^{-1} is differentiable at $f(c)$. Moreover,

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} \quad (\text{see prove below})$$

In words $(f^{-1})'(x)$ is pronounced the derivative of f inverse.



On the AP exam, inverse functions are often not expressed using inverse notation. For example, let f be a function with inverse function g then $g'(x) = \frac{1}{f'(g(x))}$.



***Teaching Tip:** Prove this formula to your students!

Definition of inverse functions: $f(f^{-1}(x)) = x$

$$\frac{d}{dx}[f(f^{-1}(x))] = \frac{d}{dx}[x]$$

Oh hey, chain rule! $f'(f^{-1}(x)) \cdot \frac{d}{dx}(f^{-1}(x)) = 1$

$$\frac{d}{dx}(f^{-1}(x)) = \frac{1}{f'(f^{-1}(x))}$$

Ex 3: Find the indicated value.

a) $f(x) = x^{5/3}$, $(f^{-1})'(243) =$

$$f(27) = 243 \Rightarrow f^{-1}(243) = 27$$

$$f'(x) = \frac{5}{3}x^{2/3}$$

$$(f^{-1})'(243) = \frac{1}{f'(f^{-1}(243))} = \frac{1}{f'(27)} = \frac{1}{15}$$

b) $f(x) = 2x + 7$, $(f^{-1})'(14) =$

$$f\left(\frac{7}{2}\right) = 14 \Rightarrow f^{-1}(14) = \frac{7}{2}$$

$$f'(x) = 2$$

$$(f^{-1})'(14) = \frac{1}{f'(f^{-1}(14))} = \frac{1}{f'\left(\frac{7}{2}\right)} = \frac{1}{2}$$

Ex 4: If $f(x) = x^5 + 2x^3 + x - 1$ and $(f \circ g)(x) = (g \circ f)(x) = x$, find $g'(3)$.

$$f(x) = 3 \Rightarrow x = 1 \text{ (solving by guess and check is quickest!)}$$

$$f(1) = 3 \Rightarrow f^{-1}(3) = g(3) = 1$$

$$f'(x) = 5x^4 + 6x^2 + 1$$

$$g'(3) = (f^{-1})'(3) = \frac{1}{f'(f^{-1}(3))} = \frac{1}{f'(1)} = \frac{1}{11}$$

Ex 5: If $f(x) = x^3 + x$ and $h(x)$ is the inverse of $f(x)$, then $h'(2)$ is

A) $\frac{1}{13}$

☒ B) $\frac{1}{4}$

C) 1

D) 4

E) 13

$$f(x) = 2 \Rightarrow x = 1$$

$$f(1) = 2 \Rightarrow f^{-1}(2) = h(2) = 1$$

$$f'(x) = 3x^2 + 1$$

$$h'(2) = (f^{-1})'(2) = \frac{1}{f'(f^{-1}(2))} = \frac{1}{f'(1)} = \frac{1}{4}$$

SCORING STATS!

Guess the percentage of students who answered correctly:

18%

Ex 6: Let f be a differentiable function such that $f(3) = 15$, $f(6) = 3$, $f'(3) = -8$ and $f'(6) = -2$. The function g is differentiable and $g(x) = f^{-1}(x)$ for all x . What is the value of $g'(3)$?

☒ A) $-\frac{1}{2}$

B) $-\frac{1}{8}$

C) $\frac{1}{6}$

D) $\frac{1}{3}$

E) The value of $g'(3)$ cannot be determined from the given information.

$$f(x) = 3 \Rightarrow x = 6$$

$$f(6) = 3 \Rightarrow f^{-1}(3) = g(3) = 6$$

$$g'(3) = (f^{-1})'(3) = \frac{1}{f'(f^{-1}(3))} = \frac{1}{f'(6)} = \frac{1}{-2}$$

SCORING STATS!

Guess the percentage of students who answered correctly:

14%



Ex 7: 2007 AB3 – part (d)

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	-4	4	2
4	-1	3	6	7

SCORING STATS!
Guess the mean: 0.96/9 🏆

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x . The function h is given by $h(x) = f(g(x)) - 6$.

- (d) If g^{-1} is the inverse function of g , write an equation for the line tangent to the graph of $y = g^{-1}(x)$ at $x = 2$.

(d) $g(1) = 2$, so $g^{-1}(2) = 1$.

$$(g^{-1})'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.

$$3 : \begin{cases} 1 : g^{-1}(2) \\ 1 : (g^{-1})'(2) \\ 1 : \text{tangent line equation} \end{cases}$$

[2007 Scoring Guidelines](#)

Ex 8: Spicy 🌶️🌶️🌶️

Suppose g is the inverse function of a differentiable function f and $G(x) = \frac{1}{g(x)}$. If $f(3) = 7$

and $f'(3) = \frac{1}{9}$, find $G'(7)$.

(A) -5

(B) 4

(C) 6

(D) -1

(E) -4

$$G(x) = \frac{1}{g(x)} = (g(x))^{-1}$$

$$G'(x) = -(g(x))^{-2} \cdot g'(x) = -\frac{g'(x)}{(g(x))^2}$$

$$G'(7) = -\frac{g'(7)}{(g(7))^2} = -\frac{9}{(3)^2} = -1$$

$$f(3) = 7 \Rightarrow f^{-1}(7) = g(7) = 3$$

$$g'(7) = \frac{1}{f'(f^{-1}(7))} = \frac{1}{f'(3)} = \frac{1}{1/9} = 9$$